Platform Verification Process

- For an SVM to start securely, three conditions must be met:
  - Platform must be verified
  - SVM authorized to run
  - SVM integrity verified
- All verification happens during the Enter Secure Mode (ESM) ultracall.
- Platform verification uses TPM and involves accessing the symmetric seed protecting the ESM operand and verifying firmware is in correct state.

Lockbox

What is a Lockbox?

A symmetric seed wrapped in a public key associated with a private key in the target system's TPM. If no lockbox exists, SVM is not authorized.

Verification Steps

TPM extracts symmetric seed from lockbox. Firmware state verified through TPM platform configuration register (PCR) 6 value.

Generating Lockbox

1 Enrollment

Extract and authenticate TPM storage key from target machine

2 Validation

Validate hardware key hash against trusted vendor manifest

3 Generation

Create symmetric seed and wrap with public key using TPM duplicate

4 Insertion

Insert lockbox into ESM operand of SVM

Local Attestation

Local attestation verifies SVM integrity using information in the ESM operand. If Ultravisor has access to symmetric seed, it generates HMAC and symmetric keys.

Generate Keys

Create HMAC key and symmetric key from seed

Verify Integrity

Check ESM operand payload block integrity

Extract Information

Use symmetric key to get integrity data

Validate Components

Verify kernel, command line, initramfs, RTAS

Hardware State Verification

- PEF relies on secure and trusted boot of the firmware stack.
- Only firmware signed by the owner (represented by hardware key hash) will be loaded.
- Secure boot verifies firmware validity, while trusted boot records information for state validation.